Production note: NovaMart Assist on Google Cloud

1. Gemini Enterprise plus custom agents

Gemini Enterprise is the front door: one assistant with single sign-on and permission-aware search, so plain policy questions need no custom code. Workflows that touch live data or money (returns, lost orders) run as **custom ADK agents on Agent Engine**, registered in Gemini Enterprise; new workflows are added the same way. Every GCP environment uses **Vertex AI** rather than an API key: service accounts, audit logs, enterprise quotas.

2. Connecting the data

Policies stay in original sources (SharePoint, Drive, Confluence) and are indexed by **Vertex AI Search**, using Google's managed parsers, layout-aware chunking and embeddings. Each chunk carries owner, version and effective date. Orders come through an API gateway, returning only the fields the agent needs; catalog and recall data come from **BigQuery**; escalations go to Help Desk. If document volume spikes, ingestion becomes event-driven (Pub/Sub, Cloud Run, Document AI).

3. Security and governance

The agent reads data and opens tickets but **never moves money**: refunds stay in the POS, and goodwill decisions stay with a manager. Personal data is kept out of tool results and redacted from logs. **Model Armor** screens prompts, retrieved passages and replies for injection, and ingestion strips hidden content from documents. Every answer is logged to BigQuery with the user, tools and versions used.

4. Quality over time

The eval suite and unit tests from this repo become a **CI release gate** for any change to prompts, models, rules or documents; return rules move to a versioned file owned by Store Ops. We track **deflection**, **draft acceptance**, **declines by topic** and **manager overrides**. Each week an LLM judge reviews sampled conversations, people check the disagreements, and failures become new eval cases.

5. Rollout

Phase 1, ticket desk (4 weeks): stores keep opening tickets, the assistant drafts replies and the ops team approves them, so we measure accuracy with no change for stores. **Phase 2, chat pilot in 3–5 stores:** once about 90% of drafts are approved unedited, then store by store.